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Loss of a metal-binding site in gelsolin leads to familial amyloidosis-Finnish type

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Abstract

Mutations in domain 2 (D2, residues 151-266) of the actin-binding protein gelsolin cause familial amyloidosis-Finnish type (FAF). These mutations, D187N or D187Y, lead to abnormal proteolysis of plasma gelsolin at residues 172-173 and a second hydrolysis at residue 243, resulting in an amyloidogenic fragment. Here we present the structure of human gelsolin D2 at 1.65 A and find that Asp 187 is part of a Cd2+ metal-binding site. Two Ca2+ ions are required for a conformational transition of gelsolin to its active form. Differential scanning calorimetry (DSC) and molecular dynamics (MD) simulations suggest that the Cd2+-binding site in D2 is one of these two Ca2+-binding sites and is essential to the stability of D2. Mutation of Asp 187 to Asn disrupts Ca2+ binding in D2, leading to instabilities upon Ca2+ activation. These instabilities make the domain a target for aberrant proteolysis, thereby enacting the first step in the cascade leading to FAF.

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